

to maximize the entire objective function. This type of decision making timeline is also exactly opposite to what happens in practice where shares are transacted first in order to arrive at the optimal portfolio. The process formulated in [Equation 10.17](#) is based on transacting shares in the market first to be consistent with practice. And addressing these issues in reverse order—e.g., determine first shares S and then trade schedule x —is only part of the whole picture and is likely to make overall performance even worse.

CONCLUSION

In this chapter we presented a process to unify the investment and trading theories. We presented a framework that overlays the efficient trading frontier (ETF) onto the efficient investment frontier to determine a set of cost-adjusted frontiers. The analysis showed that while a traditional Almgren-Chriss trade cost optimization will result in numerous efficient strategies, there is only a single “optimal” execution strategy consistent with the underlying investment objective.

The analysis also shows that a traditional VWAP strategy is not consistent with the investment objective and may compromise the portfolio manager’s stock selection ability by resulting in lower levels of investor utility. The reason is that the corresponding VWAP frontier is inferior (lies below) to the cost-adjusted frontier. Furthermore, an overly aggressive execution strategy is also an inappropriate strategy because its cost-adjusted frontier lies below the optimal cost-adjusted frontier. To maximize investor utility it is essential that the trading strategy not incur too much impact (aggressive strategy) or too much risk (VWAP strategy). Doing so is likely to result in sub-par performance.

In the last part of this chapter we presented a methodology to incorporate variable trading cost estimates (market impact and timing risk) directly into the investment optimization process. Recent attempts in this arena have been insufficient since resulting cost estimates have only been dependent upon the number of shares transacted not on the overall list composition. Managers could achieve performance improvement by incorporating market impact cost estimates directly into the investment process such that costs will be dependent upon shares transacted and trading strategy, taking advantage of overall risk reduction. This in turn could dramatically reduce the overall cost of the list.

The resulting procedure, however, is a relatively difficult non-linear multi-period optimization problem but recent advancements in